

## KIELCE-SUKOW, Poland

WMO: 125700

Lat: 50.8103N

Lon: 20.6922E

Elev: 261

StdP: 98.23

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -16.2      | -12.9 | -18.4                         | 0.8 | -15.5 | -15.6 | 1.0 | -12.3 | 10.3                  | 4.0  | 9.2 | 2.8  | 1.6                   | 70   | 0.543 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.2                   | 29.9            | 20.3 | 28.0 | 19.4 | 26.3 | 18.5 | 21.3                | 28.0 | 20.4 | 26.4 | 19.4 | 24.6 | 3.4                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.0                            | 14.3 | 23.7 | 18.2 | 13.5 | 22.8 | 17.4 | 12.9 | 21.9 | 62.9          | 28.0 | 59.6 | 26.6 | 56.4 | 24.6 | 25.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.4           | 7.3  | 6.4 | DB                         | -19.7 | 32.7               | 4.7 | 1.7                                                | -23.1 | 33.9       | -25.8 | 34.9       | -28.4 | 35.8       | -31.9 | 37.1 | (4) |
| (5)               |      |     | WB                         | -19.9 | 23.1               | 4.6 | 1.0                                                | -23.3 | 23.8       | -26.0 | 24.4       | -28.6 | 24.9       | -31.9 | 25.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 8.2         | -2.6  | -1.1  | 2.5   | 8.4   | 13.2  | 16.5  | 18.6  | 18.1  | 13.3  | 8.2   | 3.4   | -1.0  | (6)  |  |
|                                                                        | DBStd       | 8.56        | 5.31  | 5.03  | 4.39  | 4.19  | 3.79  | 3.49  | 3.23  | 3.26  | 3.56  | 4.13  | 4.28  | 4.85  | (7)  |  |
|                                                                        | HDD10.0     | 1666        | 391   | 312   | 235   | 78    | 13    | 1     | 0     | 0     | 10    | 85    | 201   | 342   | (8)  |  |
|                                                                        | HDD18.3     | 3822        | 649   | 545   | 492   | 299   | 164   | 76    | 37    | 45    | 154   | 313   | 448   | 600   | (9)  |  |
|                                                                        | CDD10.0     | 997         | 0     | 0     | 1     | 29    | 111   | 197   | 267   | 250   | 109   | 30    | 3     | 0     | (10) |  |
|                                                                        | CDD18.3     | 111         | 0     | 0     | 0     | 4     | 21    | 46    | 37    | 3     | 0     | 0     | 0     | 0     | (11) |  |
|                                                                        | CDH23.3     | 1259        | 0     | 0     | 0     | 12    | 79    | 240   | 486   | 389   | 52    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH26.7     | 321         | 0     | 0     | 0     | 1     | 8     | 55    | 141   | 109   | 7     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.8         | 3.1   | 3.2   | 3.3   | 3.0   | 2.7   | 2.6   | 2.4   | 2.2   | 2.3   | 2.6   | 2.8   | 3.0   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 631         | 37    | 34    | 41    | 40    | 69    | 72    | 99    | 65    | 55    | 41    | 43    | 35    | (15) |  |
|                                                                        | PrecMax     | 891         | 94    | 93    | 78    | 142   | 138   | 174   | 274   | 128   | 137   | 103   | 84    | 84    | (16) |  |
|                                                                        | PrecMin     | 485         | 8     | 8     | 11    | 1     | 24    | 24    | 3     | 12    | 6     | 5     | 0     | 5     | (17) |  |
|                                                                        | PrecStd     | 91          | 17    | 17    | 18    | 27    | 33    | 35    | 65    | 29    | 32    | 27    | 18    | 17    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 9.5   | 12.0  | 17.5  | 24.9  | 27.9  | 30.7  | 32.6  | 32.4  | 27.6  | 21.9  | 15.2  | 10.1  | (19) |  |
|                                                                        |             | MCWB        | 7.1   | 8.1   | 10.5  | 14.8  | 17.8  | 21.0  | 21.6  | 20.9  | 18.0  | 15.5  | 11.9  | 8.3   | (20) |  |
|                                                                        | 2%          | DB          | 7.0   | 9.2   | 14.3  | 21.6  | 25.6  | 28.2  | 30.2  | 29.4  | 24.6  | 19.5  | 13.0  | 8.1   | (21) |  |
|                                                                        |             | MCWB        | 5.4   | 6.7   | 8.6   | 13.3  | 17.0  | 19.8  | 20.1  | 20.3  | 17.2  | 14.5  | 10.6  | 6.6   | (22) |  |
|                                                                        | 5%          | DB          | 5.4   | 7.3   | 11.8  | 18.8  | 23.6  | 26.0  | 28.0  | 27.4  | 22.5  | 17.3  | 11.1  | 6.4   | (23) |  |
|                                                                        |             | MCWB        | 4.2   | 5.4   | 7.6   | 11.8  | 16.2  | 18.4  | 19.6  | 19.4  | 16.2  | 13.5  | 9.4   | 5.4   | (24) |  |
|                                                                        | 10%         | DB          | 3.9   | 5.5   | 9.4   | 16.5  | 21.3  | 24.1  | 26.1  | 25.3  | 20.4  | 14.9  | 9.7   | 4.9   | (25) |  |
|                                                                        |             | MCWB        | 2.9   | 3.9   | 6.4   | 10.5  | 15.1  | 17.5  | 18.6  | 18.3  | 15.2  | 12.3  | 8.3   | 4.1   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.6   | 8.6   | 11.1  | 15.5  | 19.6  | 22.0  | 22.6  | 22.2  | 19.2  | 16.6  | 12.3  | 8.7   | (27) |  |
|                                                                        |             | MCDB        | 8.9   | 10.7  | 15.5  | 23.3  | 25.6  | 28.7  | 30.1  | 29.4  | 25.4  | 20.4  | 14.3  | 9.8   | (28) |  |
|                                                                        | 2%          | WB          | 5.7   | 7.0   | 9.4   | 14.1  | 18.0  | 20.6  | 21.3  | 21.1  | 18.0  | 15.1  | 11.0  | 6.8   | (29) |  |
|                                                                        |             | MCDB        | 6.9   | 8.9   | 13.3  | 20.4  | 23.6  | 26.7  | 28.1  | 27.8  | 22.9  | 18.6  | 12.7  | 7.8   | (30) |  |
|                                                                        | 5%          | WB          | 4.3   | 5.5   | 8.0   | 12.5  | 16.7  | 19.3  | 20.4  | 20.1  | 16.9  | 13.8  | 9.6   | 5.5   | (31) |  |
|                                                                        |             | MCDB        | 5.3   | 7.1   | 11.1  | 17.6  | 22.2  | 24.7  | 26.4  | 26.0  | 21.4  | 16.7  | 10.9  | 6.4   | (32) |  |
|                                                                        | 10%         | WB          | 2.9   | 4.0   | 6.6   | 11.2  | 15.6  | 18.2  | 19.5  | 19.0  | 15.8  | 12.5  | 8.3   | 4.1   | (33) |  |
|                                                                        |             | MCDB        | 3.8   | 5.3   | 9.1   | 15.5  | 20.5  | 22.9  | 24.3  | 24.0  | 19.6  | 14.8  | 9.5   | 4.8   | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.4   | 6.7   | 8.5   | 10.9  | 11.3  | 11.0  | 11.2  | 11.6  | 10.2  | 8.9   | 6.1   | 5.2   | (35) |  |
|                                                                        |             | MCDBR       | 6.0   | 9.3   | 13.6  | 15.6  | 15.1  | 14.4  | 14.9  | 14.9  | 14.2  | 12.8  | 8.9   | 6.4   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 5.0   | 6.7   | 8.5   | 8.1   | 7.8   | 6.9   | 6.2   | 6.6   | 7.3   | 8.0   | 6.6   | 5.3   | (37) |  |
|                                                                        |             | MCWBR       | 5.8   | 8.5   | 12.4  | 14.0  | 13.5  | 13.0  | 12.9  | 13.5  | 12.6  | 12.0  | 7.9   | 6.1   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.317 | 0.342 | 0.384 | 0.434 | 0.423 | 0.421 | 0.445 | 0.437 | 0.403 | 0.371 | 0.349 | 0.313 | (40) |  |
|                                                                        |             | taud        | 2.249 | 2.239 | 2.197 | 2.123 | 2.210 | 2.247 | 2.194 | 2.218 | 2.273 | 2.330 | 2.320 | 2.306 | (41) |  |
|                                                                        | Edn at Noon | Edn at Noon | 686   | 771   | 804   | 804   | 833   | 837   | 808   | 793   | 777   | 731   | 645   | 635   | (42) |  |
|                                                                        |             | Edh at Noon | 71    | 93    | 117   | 141   | 135   | 131   | 137   | 127   | 108   | 85    | 67    | 58    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 0.79  | 1.54  | 2.71  | 4.08  | 5.08  | 5.45  | 5.12  | 4.60  | 3.27  | 1.92  | 0.89  | 0.60  | (44) |  |
|                                                                        | RadStd      | RadStd      | 0.14  | 0.27  | 0.35  | 0.50  | 0.57  | 0.45  | 0.61  | 0.34  | 0.48  | 0.27  | 0.10  | 0.09  | (45) |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.48 | N/A     | N/A    | +0.91   | N/A   | N/A   | N/A         | N/A     | +57     | +28     |
| (47) Regional (32 neighbors) |  | +0.51 | N/A     | N/A    | +0.83   | +0.41 | +0.37 | N/A         | -161    | +70     | +27     |

Nomenclature: See separate page